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20 Sep. 2026 Lecture 4 Boolean Satisfiability (SAT) 布尔可满足性

"Satisfy" -> find an input to make the function = 1

Most convenient form to solve this: **Conjunctive Normal Form (CNF)** (products of sum)

$$\Phi = (a + c)(b + c)(a + b + \bar{c})$$

↓ clause
↓ positive literal
→ negative literal

if a clause is 0, then the whole function is 0

after **assignment**, a clause is:

conflicting if it equals 0, **satisfied** if it equals 1, **unsolved** if you don't know its value yet

Boolean Constraint Propagation (BCP) 布尔约束传播

Unit Clause Rule: "Unit" means a clause has one unassigned literal

"Gosh, this is like 50 years old", said Ruten

$$(\neg a + \neg b + \neg c)$$

if $\begin{cases} a=1 \\ b=1 \end{cases} \rightarrow (\neg c) \rightarrow$ Deduce that: c must be 0, or it will be conflicting $\Rightarrow$ Simplifying itself

BCP iterative - Go until no more implications

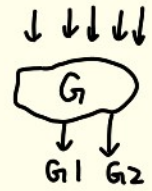
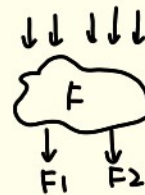
Still the heart and soul of a SAT engine: **this is called DPLL algorithms**

50,000 vars ; 25,000,000 literals ; 50,000,000 clauses

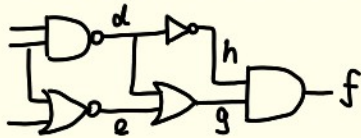
Typical SAT problems:

$F == G?$

Check: $H = F \oplus G$ if H **unsat**, then $F == G$



Then question is: How do I get CNF?



$$\phi_d = [d == (\overline{a} \overline{b})] = d \oplus (\overline{a} \overline{b})$$

$$(a+d)(b+d)(\overline{a} + \overline{b} + \overline{d}) = \phi_d$$

gate consistency function
 $==1 \rightarrow$ yes, that's what the f does
 $==0 \rightarrow$ no.

So, we can make $\phi_d, \phi_e, \phi_h, \phi_g$ $\phi_g = g \oplus (d+e) = (g+\overline{d})(\overline{e}+g)(d+e+\overline{g})$

CNF $\varphi = (\text{output var}) \prod \phi_k$

$$z = \text{NOR}(x_1, x_2, \dots, x_n) \quad z = \text{OR}(x_1, x_2, \dots, x_n)$$

$$\left[\prod_{i=1}^n (\bar{x}_i + \bar{z}) \right] \left[\left(\sum_{i=1}^n x_i \right) + z \right] \quad \left[\prod_{i=1}^n (\bar{x}_i + z) \right] \left[\left(\sum_{i=1}^n x_i \right) + \bar{z} \right]$$

$$z = \text{NAND}(x_1, x_2, \dots, x_n) \quad z = \text{AND}(x_1, x_2, \dots, x_n)$$

$$\left[\prod_{i=1}^n (x_i + z) \right] \left[\left(\sum_{i=1}^n \bar{x}_i \right) + \bar{z} \right] \quad \left[\prod_{i=1}^n (x_i + \bar{z}) \right] \left[\left(\sum_{i=1}^n \bar{x}_i \right) + z \right]$$

$z = (x)$

$$[\bar{x} + z][x + \bar{z}]$$

$\bar{z} = \text{NOT}(x)$

$$[x + z][\bar{x} + \bar{z}]$$

$\oplus, \odot$ are difficult for SAT